

# STOCK ANALYSIS MODEL

User Guide & Decision Reference

A systematic, moat-first framework for equity valuation

Updated April 2026

# Contents

---

## Chapter 1 — Introduction & Philosophy

- The ROIC > WACC principle
- Buffett / Munger framework

## Chapter 2 — Running the Pipeline

- Prerequisites & commands
- Output files
- Troubleshooting

## Chapter 3 — Universe & Quality Screen

- Ticker sources
- ROIC & WACC explained
- What the screen means

## Chapter 4 — Valuation Models

- DCF, DDM, EPV, RIM
- Monte Carlo confidence
- Composite fair value

## Chapter 5 — The Scoring System

- 5 categories & weights
- 20 gates
- Hard-gate overrides

## Chapter 6 — Ratings: BUY to PASS

- Score thresholds
- Hard gates
- Macro adjustments

## Chapter 7 — Reading the HTML Report

- Tab overview
- Detail modal
- How to build conviction

## Chapter 8 — Profit Pool Analysis

- Revenue & profit share
- pp\_multiple
- HHI & CR4

## Chapter 9 — Macro Overlay

- Indicators
- Regime classification
- Effect on valuations

## Chapter 10 — From Rating to Decision

- 7-step entry checklist
- Position sizing
- Exit criteria
- Portfolio construction
- Common mistakes

### **Appendix A — Key Metrics Quick Reference**

- All metrics with gate thresholds

### **Appendix B — Printable Master Checklist**

- One page — print and use per stock

## CHAPTER 1

# Introduction & Philosophy

## IN THIS CHAPTER

- › Why ROIC > WACC is the only screen that matters
- › How the three pipeline phases work together
- › The Buffett / Munger philosophy behind the weightings

## CORE PRINCIPLE

A business that earns ROIC > WACC creates value with every dollar it retains.  
 One that earns ROIC < WACC destroys value even when it grows.  
 The screen is a quality gate — only confirmed value creators proceed.

## What the Pipeline Does

- **Phase 1 — Screen:** Fetches 500+ tickers, computes 5-year average ROIC and WACC, retains companies where ROIC exceeds WACC. Typically 40–60% of the universe passes.
- **Phase 2 — Value:** Runs four independent valuation models (DCF, DDM, EPV, RIM) with Monte Carlo simulation, blends them into a composite fair value with confidence bounds.
- **Phase 3 — Score & Rate:** Scores each stock across five weighted categories using 20 gates, applies hard-gate overrides and macro adjustments, assigns BUY / LEAN BUY / HOLD / PASS.

## The Buffett / Munger Framework

Moat carries 40% of the composite score — far more than Valuation (20%) — because a wide, durable moat compounds value over time and makes precise valuation less critical. Overpaying slightly for a great business beats buying a mediocre one at a discount.

The model rewards: persistent ROIC > WACC spread, gross margin pricing power, cash-based earnings, insider ownership alignment, and disciplined capital return. It penalises: high leverage, earnings manipulation flags, erratic ROIC, and stock-based-compensation dilution.

**CHAPTER CHECKLIST — verify before moving on**

- ☐ I understand that  $ROIC > WACC$  means the business creates value
- ☐ I know the three phases: Screen → Value → Score & Rate
- ☐ I understand why Moat is weighted highest (40%)
- ☐ I know Financial Services stocks are excluded by design

## CHAPTER 2

# Running the Pipeline

**IN THIS CHAPTER**

- › Two commands: standard run and macro-overlay run
- › Expected runtime 15–25 minutes for the full universe
- › Three output files: JSON, HTML report, Excel workbook

## Prerequisites

- Python 3.9+ with virtual environment active
- Dependencies installed: `pip install -r requirements.txt`
- Finnhub API key set (for insider data, news, supply chain)
- Internet connection — live data fetched from Wikipedia, Yahoo Finance, SEC EDGAR, Finnhub

## Commands

### Standard run (no macro)

```
python scripts/analyze_stock.py
```

### With macro overlay — recommended for daily use

```
python scripts/analyze_stock.py --macro
```

The `--macro` flag fetches real-time VIX, yield curve, credit spreads, and sector ETF momentum, then adjusts DCF inputs and score weights to reflect the current market regime.

## Output Files

| File                                          | Description                                                                    |
|-----------------------------------------------|--------------------------------------------------------------------------------|
| output/results_YYYY-MM-DD.json                | Full results with every metric. Source of truth for the HTML report.           |
| output/stock_analysis_results_YYYY-MM-DD.html | Interactive report. All tabs, filters, and detail modals. Open in any browser. |
| output/stock_analysis_results_YYYY-MM-DD.xlsx | Sortable Excel workbook with results, sector summaries, and gate breakdown.    |
| output/run_YYYYMMDD.log                       | Full pipeline log. If the run fails, check the last 50 lines here.             |

**IF THE RUN FAILS**

Common causes: yfinance rate limiting (retry after 5 min), Wikipedia scrape failure, or network timeout. The pipeline skips individual tickers that fail rather than aborting.  
Run: `tail -50 output/run_YYYYMMDD.log` to see the specific error.

**CHAPTER CHECKLIST — verify before moving on**

- ☐ I know the difference between the standard run and --macro run
- ☐ I know where to find the HTML report and JSON results
- ☐ I know how to diagnose a failed run via the log file
- ☐ I understand the pipeline runs automatically at market close on weekdays

## CHAPTER 3

# Universe & Quality Screen

## IN THIS CHAPTER

- › The S&P; 500 + Dow Jones universe — ~500 tickers
- › ROIC measures how efficiently a business turns capital into profit
- › WACC is the minimum return needed to satisfy all capital providers

## Ticker Universe

The pipeline automatically fetches the S&P; 500 (~493 tickers) and Dow Jones (30 tickers) from Wikipedia, deduplicated to ~500–510 tickers. Financial Services are excluded — ROIC is not meaningful for banks where leverage is the business model.

## ROIC — Return on Invested Capital

### FORMULA

$ROIC = NOPAT \div \text{Invested Capital}$

$NOPAT = \text{Operating Income} \times (1 - \text{effective tax rate})$

$\text{Invested Capital} = \text{Total Equity} + \text{Total Debt} - \text{Cash}$

Computed for each of the last 5 years, then averaged to smooth cyclical distortions.

## WACC — Weighted Average Cost of Capital

### FORMULA

$WACC = (E/V) \times Re + (D/V) \times Rd \times (1 - \text{tax rate})$

$E/V = \text{Market cap} \div (\text{Market cap} + \text{Debt})$  [equity weight]

$D/V = \text{Debt} \div (\text{Market cap} + \text{Debt})$  [debt weight]

$Re = \text{Cost of equity via CAPM: } R_f + \beta \times ERP$

$Rd = \text{Interest expense} \div \text{Total debt}$

Sector floors/caps applied — e.g. Technology 8%–13%, Utilities 5%–9%.



## The Screen

### WHAT IT MEANS

A stock qualifies if 5-year average ROIC > WACC.

The size of the spread matters — it feeds the Moat scoring category.

A positive spread = the business creates value for shareholders.

A negative spread = growth destroys value, regardless of how cheap the stock looks.

### CHAPTER CHECKLIST — verify before moving on

- ☐ I know  $\text{ROIC} = \text{NOPAT} \div \text{Invested Capital}$  (5-year average)
- ☐ I know  $\text{WACC} = \text{blended cost of equity and after-tax debt}$
- ☐ I understand a positive ROIC–WACC spread = value creation
- ☐ I know why a low P/E on a negative-spread company is a value trap

## CHAPTER 4

# Valuation Models

## IN THIS CHAPTER

- › Four models run independently: DCF, DDM, EPV, RIM
- › DCF blends 6 growth signals and runs 250 Monte Carlo iterations
- › The composite fair value blends models with rules to prevent over-optimism

## 1. DCF — Discounted Cash Flow (Primary)

DCF projects free cash flow over 10 years at WACC. It is the primary model and carries the most weight in the composite fair value.

### Growth rate: a 6-signal weighted blend

| Signal                               | Weight |
|--------------------------------------|--------|
| Analyst long-term growth (~5Y)       | 30%    |
| Fundamental growth (ROE × retention) | 20%    |
| Analyst earnings growth (1Y)         | 15%    |
| Analyst revenue growth (1Y)          | 15%    |
| Revenue CAGR (3Y historical)         | 10%    |
| FCF CAGR (historical)                | 10%    |

### Two-stage projection

- **Years 1–5:** FCF grows at the blended growth rate
- **Years 6–10:** Growth fades linearly toward terminal growth
- **Terminal value:** Gordon Growth Model —  $FCF_{10} \times (1 + g) \div (WACC - g)$
- **Sector growth caps:** Technology 15%, Consumer Defensive 8%, Energy 5%

### Monte Carlo confidence (250 iterations)

| Output            | What It Means                                       |
|-------------------|-----------------------------------------------------|
| Median Fair Value | Central estimate — used as the DCF fair value       |
| P10 / P90 range   | Bear and bull fair values at 10th / 90th percentile |
| CV label          | HIGH < 20% · MEDIUM 20–40% · LOW > 40%              |

| Output        | What It Means                            |
|---------------|------------------------------------------|
| Score penalty | LOW confidence → -15% on composite score |

## 2. DDM — Dividend Discount Model

Applied to dividend-paying companies with  $\geq 3$  consecutive years of dividends. Anchors to future dividends rather than FCF — natural fit for utilities, REITs, stable consumer staples. When eligible: **final FV = 70% DCF + 30% DDM**.

## 3. EPV — Earnings Power Value

**EPV = NOPAT ÷ WACC + Excess Cash per share**. A zero-growth floor — the fair value assuming earnings are maintained but never grow. If the stock trades *below* EPV, the market is pricing in earnings decline. Use it as a DCF sanity check.

## 4. RIM — Residual Income Model

Starts from book value and adds the PV of returns earned above the cost of equity. More anchored than DCF when tangible assets are significant.

## Composite Fair Value

**Margin of Safety (MoS) = (Fair Value – Price) ÷ Fair Value**. MoS > 10% is a hard gate requirement. The detail modal shows the model blend used (e.g. "70% DCF + 30% DDM") and any divergence flags.

### CHAPTER CHECKLIST — verify before moving on

- ☐ I know DCF blends 6 growth signals — analyst LT growth is weighted highest (30%)
- ☐ I understand Monte Carlo confidence: HIGH < 20% CV, LOW > 40% CV
- ☐ I know EPV is a zero-growth floor — if price < EPV, market expects decline
- ☐ I know MoS = (Fair Value – Price) ÷ Fair Value, gate threshold is > 10%
- ☐ I know DDM eligible stocks get 70% DCF + 30% DDM blended fair value

## CHAPTER 5

# The Scoring System

## IN THIS CHAPTER

- › 5 weighted categories produce a composite score from 0 to 100
- › Moat gets 40% — durability of returns matters most
- › 20 individual gates each score 0–100 continuously (not binary)

## Category Weights

| Category  | Weight | What It Measures                                             |
|-----------|--------|--------------------------------------------------------------|
| Moat      | 40%    | ROIC spread, gross margin, FCF margin, ROIC consistency      |
| Quality   | 20%    | Cash conversion, leverage, earnings quality, bankruptcy risk |
| Ownership | 15%    | Insider ownership, buybacks, dividends, SBC dilution         |
| Valuation | 20%    | Margin of safety, Price/FV ratio, Price/FCF                  |
| Growth    | 5%     | Revenue CAGR, ROE, margin trend, fundamental growth          |

## WHY MOAT IS WEIGHTED 40%

Moat durability is the dominant driver of long-term equity returns.  
 A wide moat compounds value over time and provides safety that no P/E multiple can replicate.  
 Growth is weighted only 5% — growth without a moat destroys value at scale.

## Key Gates by Category

| Category | Gate                  | Hard Threshold             |
|----------|-----------------------|----------------------------|
| Moat     | ROIC–WACC Spread      | > 7% to avoid PASS rating  |
| Moat     | ROIC Consistency (CV) | < 30% to avoid PASS rating |
| Moat     | Gross Margin          | > 35% to avoid PASS rating |
| Moat     | FCF Margin            | > 12% for full gate pass   |
| Quality  | Interest Coverage     | > 3× for gate pass         |
| Quality  | Net Debt / EBITDA     | < 1.5× for gate pass       |
| Quality  | Cash Conversion       | > 85% for gate pass        |

| Category  | Gate              | Hard Threshold                     |
|-----------|-------------------|------------------------------------|
| Quality   | Accruals (Sloan)  | < 8% for gate pass                 |
| Ownership | Shareholder Yield | > 2% for gate pass                 |
| Ownership | Insider Ownership | > 5% for gate pass                 |
| Valuation | Margin of Safety  | > 10% — hard gate for BUY/LEAN BUY |

#### COMPOSITE SCORE FORMULA

Per-gate score = continuous 0–100 based on metric value

Category score = average of all gates in that category

Composite =  $\Sigma$  (category\_score  $\times$  weight)

Penalty =  $\times 0.85$  if MC confidence LOW (CV > 40%)

#### CHAPTER CHECKLIST — verify before moving on

- ☐ I know the 5 categories and their weights (Moat 40% is highest)
- ☐ I know scores are continuous 0–100, not binary pass/fail
- ☐ I know the three Moat hard gates: spread > 7%, CV < 30%, gross margin > 35%
- ☐ I know a LOW Monte Carlo confidence applies a 15% composite score penalty

## CHAPTER 6

# Ratings: BUY to PASS

## IN THIS CHAPTER

- › Four ratings: BUY ( $\geq 60$ ), LEAN BUY (40–59), HOLD ( $< 40$ ), PASS (gate fail)
- › Hard gates can downgrade any rating — they never upgrade
- › Macro overlay adjusts WACC and growth inputs before scoring

## Score Thresholds

**BUY**

**Score  $\geq 60$**  — Strong across all weighted categories. Margin of safety present, moat confirmed, quality gates pass. Conviction idea.

**LEAN BUY**

**Score 40 – 59** — Solid but not exceptional. Meaningful strength in moat or quality offset by weakness elsewhere. Monitor before adding size.

**HOLD**

**Score  $< 40$**  — Multiple red flags or moderate concerns. Failed a gate or composite too low. Hold existing positions; do not add.

**PASS**

**Score Gate fail** — Failed a critical hard gate. Typically a moat or quality failure. Excluded from consideration regardless of price.

## Hard Gates That Override Ratings

| Gate                | Threshold     | Effect                                       |
|---------------------|---------------|----------------------------------------------|
| Margin of Safety    | $< 10\%$      | Cannot be BUY or LEAN BUY $\rightarrow$ HOLD |
| Interest Coverage   | $< 3\times$   | Downgrade one level                          |
| Accruals            | $> 8\%$       | Downgrade one level                          |
| Net Debt / EBITDA   | $> 1.5\times$ | Downgrade one level                          |
| Shareholder Yield   | $< 2\%$       | Downgrade one level                          |
| ROIC Consistency CV | $> 30\%$      | PASS — moat not durable                      |
| ROIC–WACC Spread    | $< 7\%$       | PASS — narrow moat                           |
| Gross Margin        | $< 35\%$      | PASS — no pricing power                      |
| Gates passed        | $< 12/20$     | HOLD — system-wide weakness                  |

**RATINGS ONLY DOWNGRADE — NEVER UPGRADE**

The gate override system is one-directional.

Gates can reduce BUY → LEAN BUY → HOLD, but never elevate HOLD → BUY.

The composite score is the ceiling; hard gates can only lower the floor.

**CHAPTER CHECKLIST — verify before moving on**

- ☐ I know BUY =  $\geq 60$ , LEAN BUY = 40–59, HOLD =  $< 40$ , PASS = gate fail
- ☐ I know hard gates downgrade ratings but never upgrade them
- ☐ I know MoS  $< 10\%$  prevents BUY or LEAN BUY regardless of composite score
- ☐ I know ROIC CV  $> 30\%$ , spread  $< 7\%$ , or gross margin  $< 35\%$  forces PASS

## CHAPTER 7

# Reading the HTML Report

## IN THIS CHAPTER

- › Four tabs: Overview, Screening Matrix, Profit Pool, Detailed Results
- › Click any stock row to open a full detail modal
- › Insider Activity, narrative, and all 20 gate scores are inside the modal

## Report Tabs

| Tab              | What You Find There                                                                                |
|------------------|----------------------------------------------------------------------------------------------------|
| Overview         | Summary stats: qualifying count, avg ROIC-WACC spread, macro regime, BUY/LEAN BUY counts.          |
| Screening Matrix | All rated stocks with all 20 gate values, colour-coded pass/fail. Filter by rating, sector, score. |
| Profit Pool      | Sector-by-sector analysis: HHI, CR4, CR4 company cards, company highlights, bar chart.             |
| Detailed Results | Full sortable results table. Click any row for the detail modal.                                   |

## Inside the Detail Modal

| Section / Tab    | What to Look For                                                                         |
|------------------|------------------------------------------------------------------------------------------|
| Valuation panel  | DCF FV, MoS (green >30%, amber 10–30%, red <10%), P10/P90 range, model blend note        |
| MC Confidence    | HIGH / MEDIUM / LOW label. If LOW, check P10 — is it below current price?                |
| Narrative        | 5–7 financial summary bullets, tailwinds (green), headwinds (amber/red), thesis-breakers |
| Insider Activity | Buy ratio last 90 days, open-market purchases, net shares/dollars, recent transactions   |
| Fundamentals     | ROIC by year — look for consistency, not just the average                                |
| Gate Performance | All 20 gates with actual value, pass/fail, and per-gate score 0–100                      |
| Culture / Supply | Glassdoor rating, revenue per employee, key customers/suppliers                          |

## HOW TO BUILD CONVICTION ON A BUY-RATED STOCK

1. Check MoS is comfortably above 10% — ideally > 25%.
2. Verify Monte Carlo confidence is HIGH or MEDIUM.
3. Read the insider activity tab — buy ratio > 60% in 90 days is a strong confirming signal.
4. Scan the narrative headwinds for anything you can verify independently.
5. Compare DCF FV to EPV — if DCF is >2× EPV, confirm the moat supports the growth assumption.



**CHAPTER CHECKLIST — verify before moving on**

- ☐ I know where to find the detail modal (click any stock row)
- ☐ I know MoS colour coding: green > 30%, amber 10–30%, red < 10%
- ☐ I know to check insider buy ratio in the Insider Activity tab
- ☐ I know to compare DCF FV to EPV as a growth assumption sanity check
- ☐ I can find all 20 gate scores in the Gate Performance tab

## CHAPTER 8

# Profit Pool Analysis

## IN THIS CHAPTER

- › Shows where profits concentrate within each sector
- ›  $pp\_multiple > 1.0$  means a company earns more profit than its revenue share implies
- › HHI and CR4 measure market concentration — high = fewer competitors, rational pricing

## Core Metrics

| Metric           | Formula                                                 | What It Tells You                                                                                        |
|------------------|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Revenue Share    | $\text{Company revenue} \div \text{Sector revenue}$     | What fraction of the sector's top line this company owns                                                 |
| Profit Share     | $\text{Company op income} \div \text{Sector op income}$ | What fraction of the sector's profits this company captures                                              |
| PP Multiple      | $\text{Profit Share} \div \text{Revenue Share}$         | $1.0 = \text{earns its share}$ . $> 1.0 = \text{moat (over-earning)}$ . $< 1.0 = \text{under-earning}$ . |
| Margin Advantage | $\text{Company OM} - \text{Sector median OM}$           | Operating margin vs sector peers. Positive = pricing power.                                              |

## HHI and CR4

| Metric        | Interpretation                                                         |
|---------------|------------------------------------------------------------------------|
| HHI < 0.10    | Fragmented — competitive, price wars possible                          |
| HHI 0.10–0.25 | Moderate concentration — some pricing discipline                       |
| HHI > 0.25    | Concentrated oligopoly — rational pricing, lower competitive intensity |
| CR4 > 60%     | Top 4 companies control the sector — high barriers to entry            |

## Reading the Profit Pool Chart

- **Bar width** = revenue share — wider bars are bigger companies
- **Bar height** = operating margin — taller bars are more profitable per dollar of sales
- **Bar colour** = rating (green BUY/LEAN BUY, amber HOLD, red PASS)
- **Ideal**: tallest bar in the widest position — scale plus pricing power
- **Click any bar** to open that company's full detail modal

**CHAPTER CHECKLIST — verify before moving on**

- ☐ I know  $pp\_multiple > 1.0$  means a company earns more profit than its revenue share implies
- ☐ I know  $HHI > 0.25$  means a concentrated sector with rational pricing
- ☐ I can read the profit pool chart: width = revenue share, height = operating margin
- ☐ I know to compare CR4 companies against Company Highlights to find scale-and-moat leaders

## CHAPTER 9

# Macro Overlay

## IN THIS CHAPTER

- › Run with --macro to apply real-time market regime adjustments
- › Four indicators: VIX, yield curve, credit spreads, SPY momentum
- › Contraction regime raises WACC and trims growth — a BUY that survives is higher-conviction

## Indicators

| Indicator                    | Bullish / Bearish                                                |
|------------------------------|------------------------------------------------------------------|
| VIX < 15                     | Complacency — neutral. VIX > 30 = fear → raise ERP, tighten WACC |
| Yield curve (10Y–2Y) > +1.5% | Expansion. < 0% = inverted → recession risk                      |
| Credit spreads < 0.5%        | Confidence → bullish. > 1.5% = stress → raise WACC               |
| SPY / 200-day SMA > 1.05     | Uptrend → expansion. < 0.95 = downtrend → contraction            |
| Sector ETF momentum          | Which sectors have relative strength vs SPY (3M return)          |

## What Each Regime Does to the Model

| Regime      | Adjustments Applied                                                     |
|-------------|-------------------------------------------------------------------------|
| Expansion   | +0.5pp terminal growth; slightly more weight on analyst consensus       |
| Contraction | +0.5pp ERP (higher WACC); exit multiple –10%; analyst LT weight reduced |
| Late Cycle  | Growth sigma ×1.5 — widens Monte Carlo uncertainty bands                |
| Neutral     | No adjustments applied                                                  |

## PRACTICAL GUIDANCE

Check the macro regime in the report header before acting on any rating.  
 The composite score already reflects macro adjustments — no manual adjustment needed.  
 A BUY in contraction (where WACC is raised and growth is trimmed) is a higher-conviction signal than the same BUY in expansion, where the model is more generous with assumptions.

**CHAPTER CHECKLIST — verify before moving on**

- ☐ I know the four macro indicators and what each signals
- ☐ I know contraction raises WACC and trims growth assumptions
- ☐ I know the composite score already reflects macro adjustments
- ☐ I know to check the regime label in the report header before acting

## CHAPTER 10

# From Rating to Decision

Use this chapter as your repeatable decision process

## IN THIS CHAPTER

- › Work the 7-step checklist in order — an early stop means wait, not skip
- › Position size is determined by rating, confidence, macro, and insider activity combined
- › Define exit conditions before you buy — remove emotion from future decisions

The model surfaces evidence — it does not make decisions. A BUY rating means the stock scores well across moat, quality, valuation, and ownership at current prices. Work through the checklist below before acting on any rating.

## Step 1 — Confirm the Rating Is Actionable

### STEP 1 CHECKLIST

- ☐ Rating is BUY or LEAN BUY (not HOLD or PASS)
- ☐ Run is less than 3 days old  
Older runs may not reflect recent price moves or earnings releases.
- ☐ Note whether this is a new rating or persistent across  $\geq 3$  runs  
A persistent BUY-rated stock carries more weight than a first-time rating.
- ☐ Record the actual composite score, not just the label  
LEAN BUY at 58 is very different from LEAN BUY at 41.

### IF HOLD OR PASS

Do not initiate. If you already own it, skip to Exit Criteria.  
A PASS stock that looks cheap on P/E is a value trap —  $ROIC < WACC$  means the business destroys capital at scale. The screen exists for this reason.

## Step 2 — Check Margin of Safety

### STEP 2 CHECKLIST

- ☐ MoS  $> 10\%$  (hard gate minimum)

- ☐ MoS > 25% — comfortable cushion for a full position  
MoS 10–20% is thin; a 10% FV revision could eliminate your discount.
- ☐ Compare DCF fair value to EPV in the detail modal  
If they are close, MoS does not depend on heroic growth assumptions — more reliable.
- ☐ Check MoS trend across recent runs  
Shrinking MoS (price rising faster than FV revisions) is a warning even before hitting 10%.

### Step 3 — Check Monte Carlo Confidence

#### STEP 3 CHECKLIST

- ☐ Confidence is HIGH (CV < 20%) or MEDIUM (CV 20–40%)
- ☐ Look at the P10/P90 range in the detail modal  
If P10 (bear case) is below current price, the model sees a realistic loss scenario.
- ☐ If LOW confidence: require MoS > 25% as a larger buffer  
Cap position at half the normal size until confidence improves.
- ☐ HIGH confidence + wide MoS beats LOW confidence + very wide MoS  
Certainty of a moderate return is often better than uncertainty of a large one.

### Step 4 — Check the Macro Regime

#### STEP 4 CHECKLIST

- ☐ Note the regime label in the report header (Expansion / Neutral / Contraction)
- ☐ If Contraction: require MoS > 20% and reduce position size by 30%  
The model has already raised WACC and trimmed growth — apply extra caution.
- ☐ Check the composite score: 0.0 = bearish, 0.5 = mixed, 1.0 = bullish  
A score near 0.5 is genuinely mixed — do not treat it as expansion or contraction.
- ☐ Check sector ETF momentum in the macro panel  
A BUY in a sector with negative relative strength carries more risk.

### Step 5 — Check Insider Activity

#### STEP 5 CHECKLIST

- ☐ Open the Insider Activity tab in the detail modal
  - Focus on open-market purchases (Form 4 type "P") — insiders spending their own cash.
- ☐ Buy ratio > 60% in last 90 days: confirming signal
  - Clustered buying (multiple insiders simultaneously) is the strongest signal of all.
- ☐ Buy ratio < 30%: caution — reduce intended size by 20%
  - Insiders know the business better than any model. Heavy selling without a clear reason warrants scrutiny.
- ☐ Heavy selling is not always bearish
  - Executives routinely diversify large concentrated positions. Context matters.

## Step 6 — Verify the Core Moat Gates

### STEP 6 CHECKLIST

- ☐ ROIC–WACC spread > 10% (comfortably above the 7% hard gate)
  - A spread just above 7% passed the gate but barely. Size more conservatively.
- ☐ ROIC consistency CV < 20% (well below the 30% hard gate)
  - In the Fundamentals tab, review ROIC year-by-year — look for stability, not just the average.
- ☐ Gross margin is stable or expanding
  - A shrinking gross margin trend is a moat erosion warning, even if the level still passes.
- ☐ Can you articulate the qualitative moat source?
  - Switching costs, network effects, cost advantage, intangible assets — if you cannot name it, the score may be overstating durability.

## Step 7 — Size the Position and Record Your Thesis

### STEP 7 CHECKLIST

- ☐ Apply the position size from the table below
- ☐ Write your thesis in one paragraph before buying
  - Why does this business have a durable moat? What is the path to fair value? What would change your mind?
- ☐ Record: entry date, fair value, MoS, composite score, macro regime
  - These become your benchmarks on every future re-run.
- ☐ Define exit conditions in advance
  - e.g. "I will sell if MoS falls below 0%, ROIC spread falls below 5%, or a thesis-breaker materialises."
- ☐ Set a calendar reminder for the next quarterly earnings re-evaluation



## Position Sizing Guide

Suggested upper bounds for a 15–25 name portfolio. Adjust downward for sector overlap or personal risk tolerance.

| Scenario                                           | Max Size | Rationale                                                  |
|----------------------------------------------------|----------|------------------------------------------------------------|
| BUY · HIGH confidence · Expansion · Insider buying | 5–6%     | All signals aligned. Maximum conviction.                   |
| BUY · HIGH confidence · Neutral regime             | 4–5%     | Strong score; no macro headwind.                           |
| BUY · MEDIUM confidence                            | 3–4%     | Uncertainty warrants a smaller initial stake.              |
| BUY · LOW confidence                               | ≤ 2%     | Treat as speculative starter — size to afford being wrong. |
| LEAN BUY · HIGH or MEDIUM confidence               | 2–3%     | Diversification slot, not a concentration bet.             |
| LEAN BUY · LOW confidence                          | ≤ 1%     | Weak signal with model uncertainty. Monitor only.          |
| Any rating · Contraction regime                    | –30%     | Apply to any of the above. Extra caution warranted.        |

## Exit Criteria

Exits are triggered by thesis changes, not price moves alone. A price drop that widens MoS may be a buy signal. A price rally that eliminates MoS while the moat remains intact is a trim signal.

### EXIT / TRIM TRIGGERS — review these on each re-run

- ☐ Rating drops to HOLD on two consecutive runs  
One-run downgrade may be data noise — wait for confirmation before full exit.
- ☐ MoS falls below 0% (price exceeds fair value)  
Trim to half position. Stay small until re-rated or price corrects.
- ☐ ROIC–WACC spread collapses below 3%  
The core moat thesis has weakened. The screen would now reject this stock — consider full exit.
- ☐ ROIC consistency CV spikes above 40%  
Moat durability is in question. Earnings have become erratic.
- ☐ Beneish M-Score manipulation flag appears  
Non-negotiable. Exit immediately. Do not wait for confirmation.
- ☐ A thesis-breaker risk materialises  
Structural change (antitrust, platform disruption, commodity cycle peak) invalidates the DCF.
- ☐ Better-rated opportunity in same sector  
Rotating from LEAN BUY to BUY in the same sector reduces opportunity cost without adding concentration.

## Portfolio Construction Rules

### PORTFOLIO CHECKLIST

- ☐ 15–25 positions total  
Fewer than 15 = concentrated single-stock risk. More than 25 = index-like with more work.
- ☐ Maximum 25% in any one sector  
The model may produce many BUYs in a favoured sector — cap it to avoid sector bets.
- ☐ BUYs anchor (60–70% of capital); LEAN BUYs diversify  
Core positions in BUY-rated HIGH-confidence names. LEAN BUYs fill smaller slots.
- ☐ No HOLD or PASS initiations — ever  
Initiating a HOLD because "it looks cheap on P/E" undercuts the entire framework.
- ☐ In Contraction: hold 10–20% cash rather than forcing LEAN BUY positions  
The model produces fewer BUYs in contraction — trust that signal.
- ☐ Re-evaluate all holdings against the current run every quarter

## Common Mistakes

### **X Buying a PASS stock because it looks cheap on P/E**

ROIC < WACC means the business destroys capital at scale. A low P/E is a value trap, not a bargain.

### **X Ignoring the confidence label**

A BUY with LOW confidence means the model is uncertain — the valuation is sensitive to small changes in growth or WACC. Size accordingly.

### **X Acting on a stale run**

Fair value changes with earnings, analyst revisions, and macro conditions. Always act on the most recent run.

### **X Anchoring to a previous fair value**

If the model revises FV down 15%, do not assume the previous value was more correct. The new value incorporates updated data.

### **X Over-concentrating in the current cycle's winners**

Sector caps exist for this reason. Technology dominating the BUY list in an expansion is a cycle signal, not a thesis.

### **X Using the model as a trading signal**

The DCF is a multi-year valuation framework. MoS of 35% can persist for 12–18 months. Size for conviction, not for a quick re-rating.

#### **THE MODEL IS A SECOND OPINION, NOT A VERDICT**

Run the model. Read the output. Then ask: do I understand this business well enough to own it through a difficult quarter?

The composite score tells you the model's view of the evidence.

Your job is to decide whether you agree — and whether you can hold conviction when the market temporarily disagrees.

## CHAPTER A

# Key Metrics Quick Reference

Every metric in the detail modal with gate thresholds

| Metric              | Definition                                      | Gate Threshold       |
|---------------------|-------------------------------------------------|----------------------|
| DCF Fair Value      | PV of all future FCFs at WACC                   | —                    |
| Margin of Safety    | $(FV - Price) \div FV$                          | > 10%                |
| P / Fair Value      | Price $\div$ DCF Fair Value                     | < 1.0 ideal          |
| P / FCF             | Market cap $\div$ Free Cash Flow                | < 25× preferred      |
| EV / EBITDA         | Enterprise Value $\div$ EBITDA                  | vs sector median     |
| ROIC                | NOPAT $\div$ Invested Capital (5Y avg)          | > WACC               |
| ROIC–WACC Spread    | ROIC – WACC                                     | > 7%                 |
| ROIC Consistency CV | Std dev of annual ROIC $\div$ mean ROIC         | < 30%                |
| Gross Margin        | $(Revenue - COGS) \div Revenue$                 | > 35%                |
| FCF Margin          | Free Cash Flow $\div$ Revenue                   | > 12%                |
| Cash Conversion     | FCF $\div$ Operating Income                     | > 85%                |
| Piotroski F-Score   | 9-point financial health checklist (0–9)        | $\geq 7$ = improving |
| Altman Z-Score      | Bankruptcy predictor                            | > 2.99 = safe        |
| Beneish M-Score     | Earnings manipulation flag                      | No flag              |
| Interest Coverage   | EBIT $\div$ Interest Expense                    | > 3×                 |
| Net Debt / EBITDA   | $(Total Debt - Cash) \div EBITDA$               | < 1.5×               |
| Accruals            | $(Net Income - Operating CF) \div Assets$       | < 8%                 |
| Insider Ownership   | % shares held by officers & directors           | > 5%                 |
| Insider Buy Ratio   | Buys $\div$ (Buys + Sells) last 90 days         | > 60% = tailwind     |
| Shareholder Yield   | $(Dividends + Buybacks) \div Market Cap$        | > 2%                 |
| SBC Dilution        | Stock-based compensation $\div$ Revenue         | < 3%                 |
| Short Interest      | % of float held short                           | > 10% = caution      |
| Revenue CAGR (10Y)  | Compound annual revenue growth over 10 years    | > 2%                 |
| PP Multiple         | Profit Share $\div$ Revenue Share within sector | > 1.0 = moat signal  |
| PP Margin Advantage | Company op margin – sector median op margin     | > 0pp                |

| Metric        | Definition                             | Gate Threshold |
|---------------|----------------------------------------|----------------|
| MC Confidence | CV: HIGH <20%, MEDIUM 20–40%, LOW >40% | HIGH or MEDIUM |

## APPENDIX B

# Master Entry Checklist

Print one copy per stock. Work through each step in order.

Ticker: \_\_\_\_\_ Company: \_\_\_\_\_ Date: 2026-04-18 Run date: \_\_\_\_\_

| STEP 1 — RATING                                                    | STEP 4 — MACRO REGIME                                              |
|--------------------------------------------------------------------|--------------------------------------------------------------------|
| <input type="checkbox"/> BUY or LEAN BUY (not HOLD / PASS)         | <input type="checkbox"/> Regime: Expansion / Neutral / Contraction |
| <input type="checkbox"/> Run is less than 3 days old               | <input type="checkbox"/> Composite score (0–1): _____              |
| <input type="checkbox"/> Composite score: _____                    | <input type="checkbox"/> Sector ETF momentum: pos / neg            |
| <input type="checkbox"/> Persistent ( $\geq 3$ runs) or new? _____ | <input type="checkbox"/> If Contraction: MoS > 20% confirmed       |

  

| STEP 2 — MARGIN OF SAFETY                                 | STEP 5 — INSIDER ACTIVITY                                   |
|-----------------------------------------------------------|-------------------------------------------------------------|
| <input type="checkbox"/> MoS > 10% (value: _____)         | <input type="checkbox"/> Buy ratio (90 days): _____         |
| <input type="checkbox"/> MoS > 25% for full position      | <input type="checkbox"/> Buy ratio > 60%: confirming signal |
| <input type="checkbox"/> DCF FV vs EPV checked            | <input type="checkbox"/> Buy ratio < 30%: reduce size 20%   |
| <input type="checkbox"/> MoS trending: up / stable / down | <input type="checkbox"/> Open-market purchases present      |

  

| STEP 3 — MC CONFIDENCE                                | STEP 6 — MOAT GATES                                          |
|-------------------------------------------------------|--------------------------------------------------------------|
| <input type="checkbox"/> Confidence: HIGH / MED / LOW | <input type="checkbox"/> ROIC-WACC spread > 10%: _____       |
| <input type="checkbox"/> CV value: _____              | <input type="checkbox"/> ROIC CV < 20%: _____                |
| <input type="checkbox"/> P10 bear case: _____         | <input type="checkbox"/> Gross margin stable or expanding    |
| <input type="checkbox"/> P10 is above current price   | <input type="checkbox"/> Can articulate the qualitative moat |

  

| STEP 7 — POSITION SIZE & THESIS                                                                               |
|---------------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> Position size determined: _____ (ref: sizing guide in Chapter 10)                    |
| <input type="checkbox"/> Thesis written (one paragraph): moat source, path to FV, what would change your mind |
| <input type="checkbox"/> Entry benchmarks recorded: FV _____ · MoS _____ · Score _____ · Regime _____         |

- ☐ Exit conditions pre-committed: MoS < 0%, spread < 5%, thesis-breaker materialises
- ☐ Quarterly review reminder set

**NOTES**

---

---

---

---

---